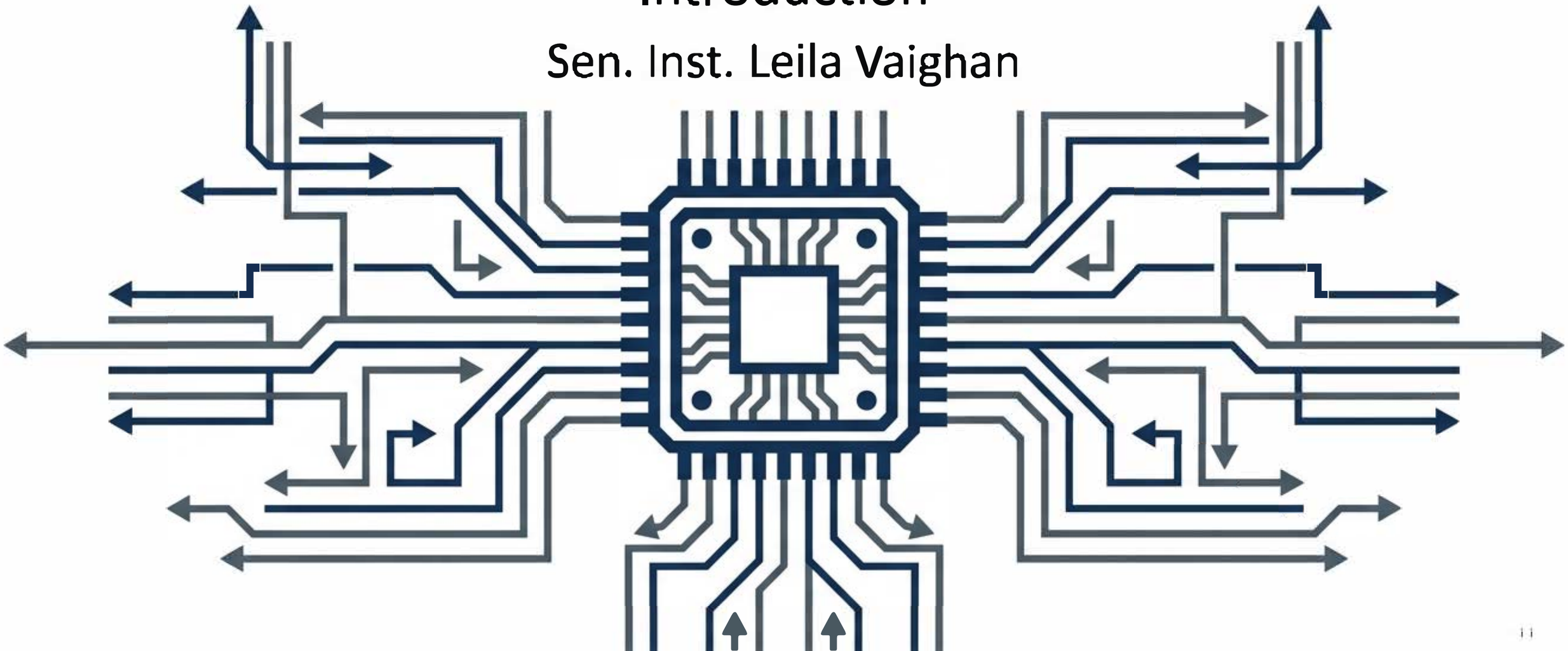
Microprocessors

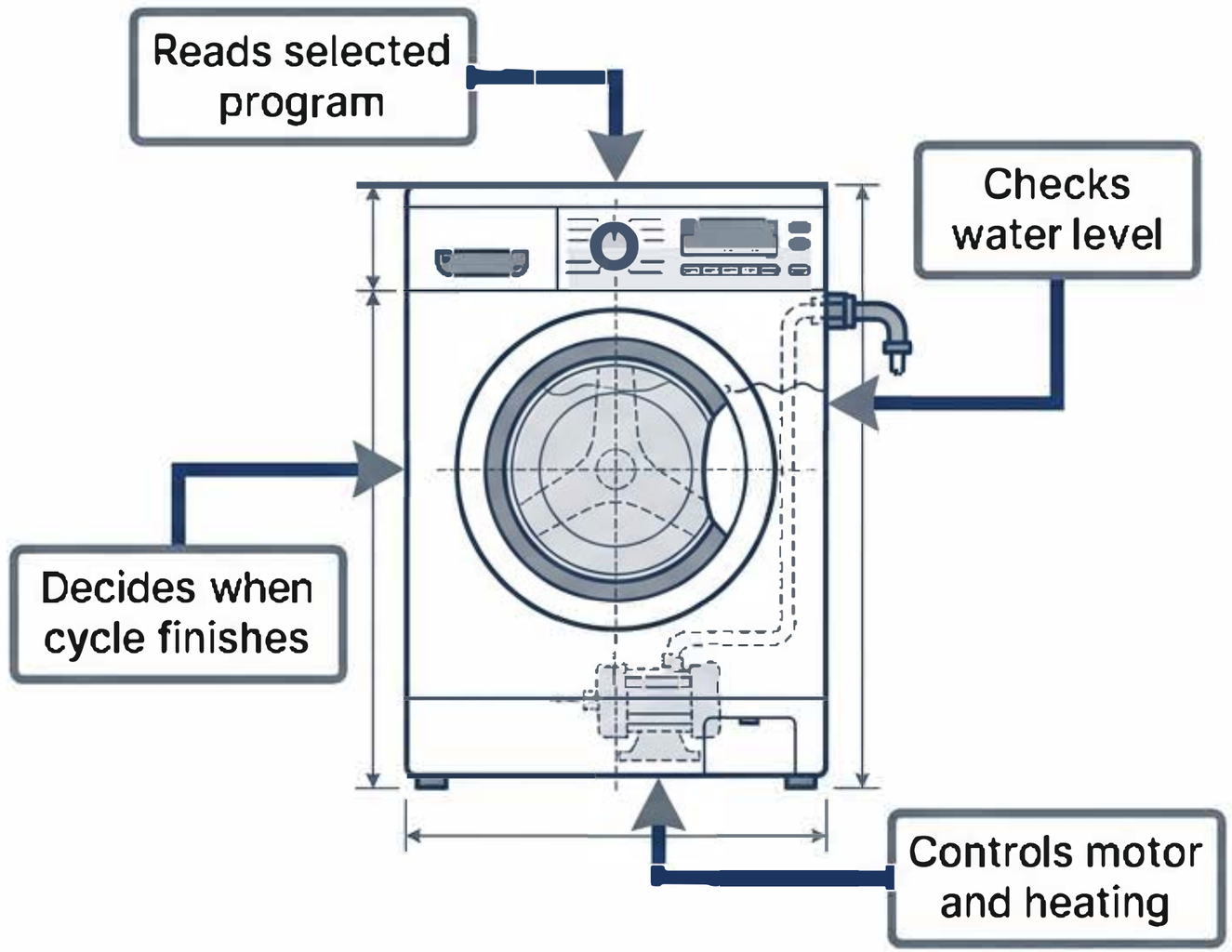
Introduction

Sen. Inst. Leila Vaighan

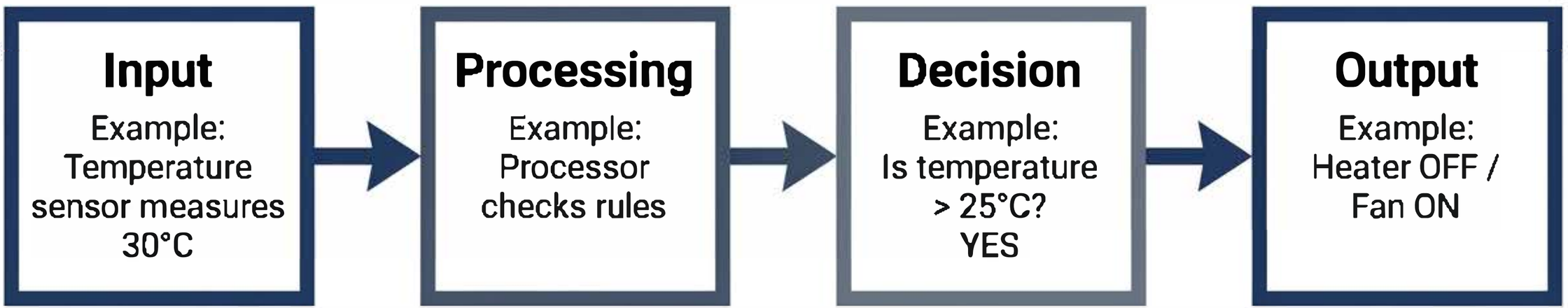


What is a Processor?

A processor is an electronic device that executes instructions to process information, make decisions, and control devices.

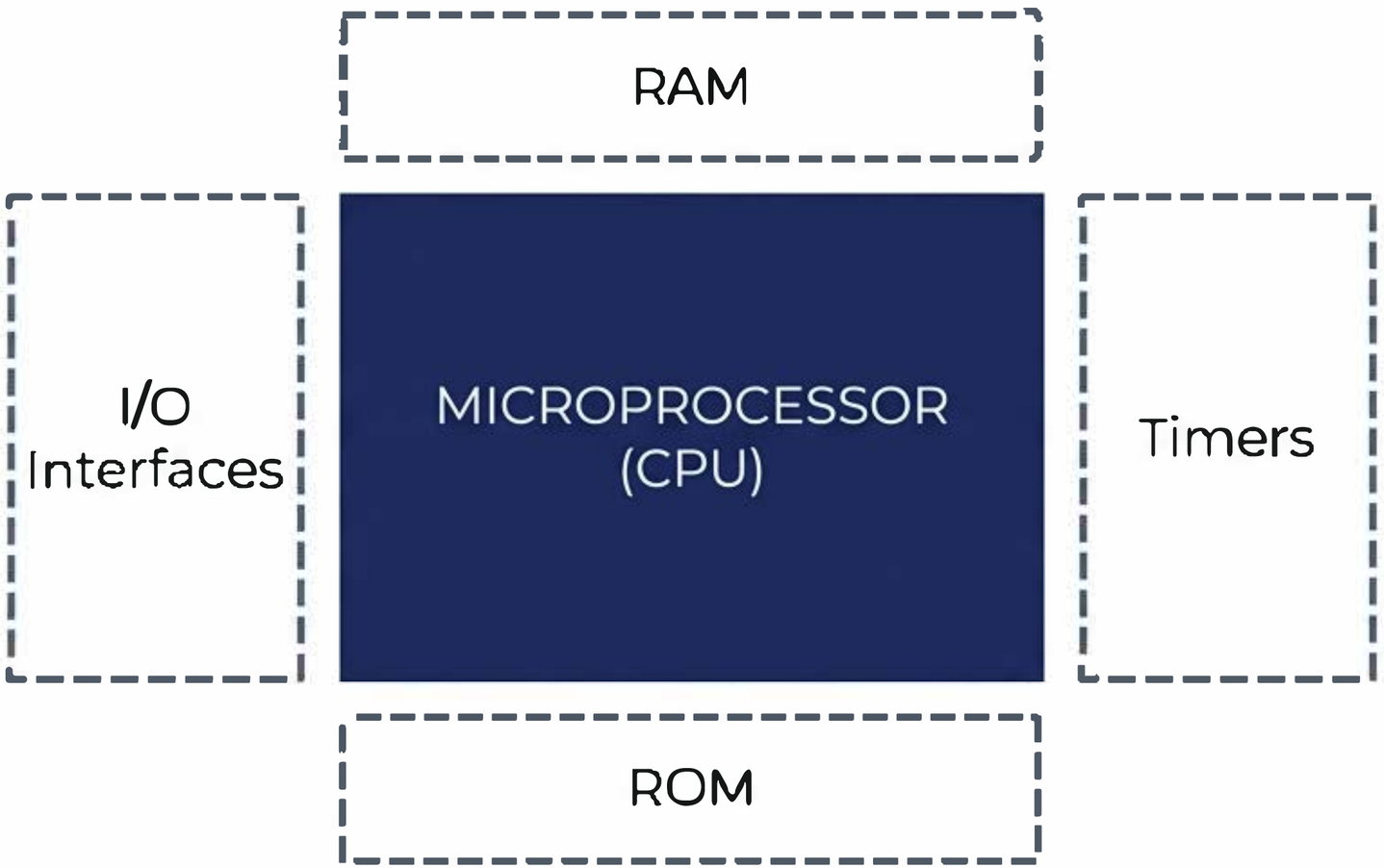


The Universal Formula



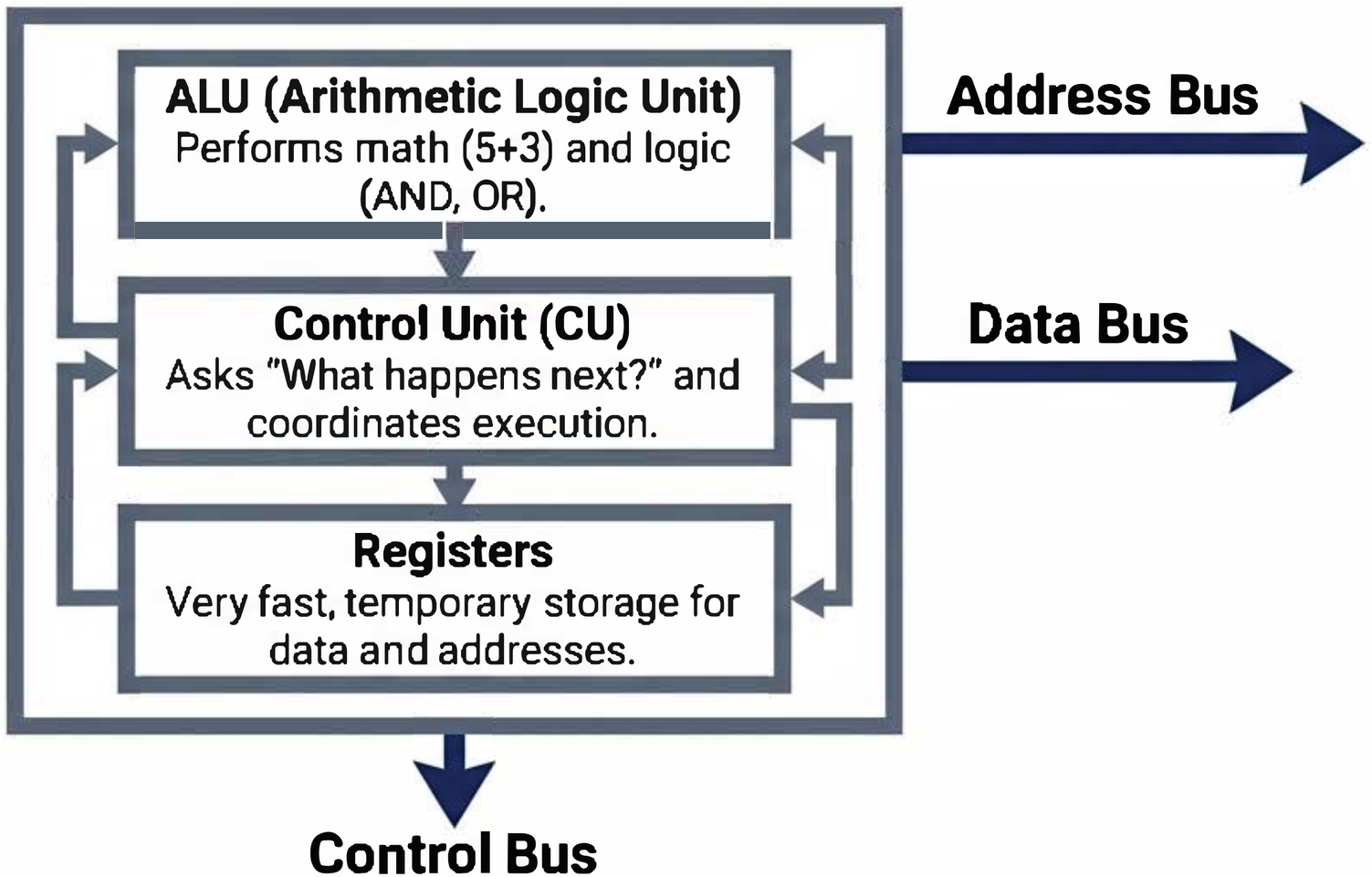
Defining the Microprocessor

A Central Processing Unit (CPU) implemented on a single integrated circuit.
It acts as the brain of a system, but it cannot function alone.



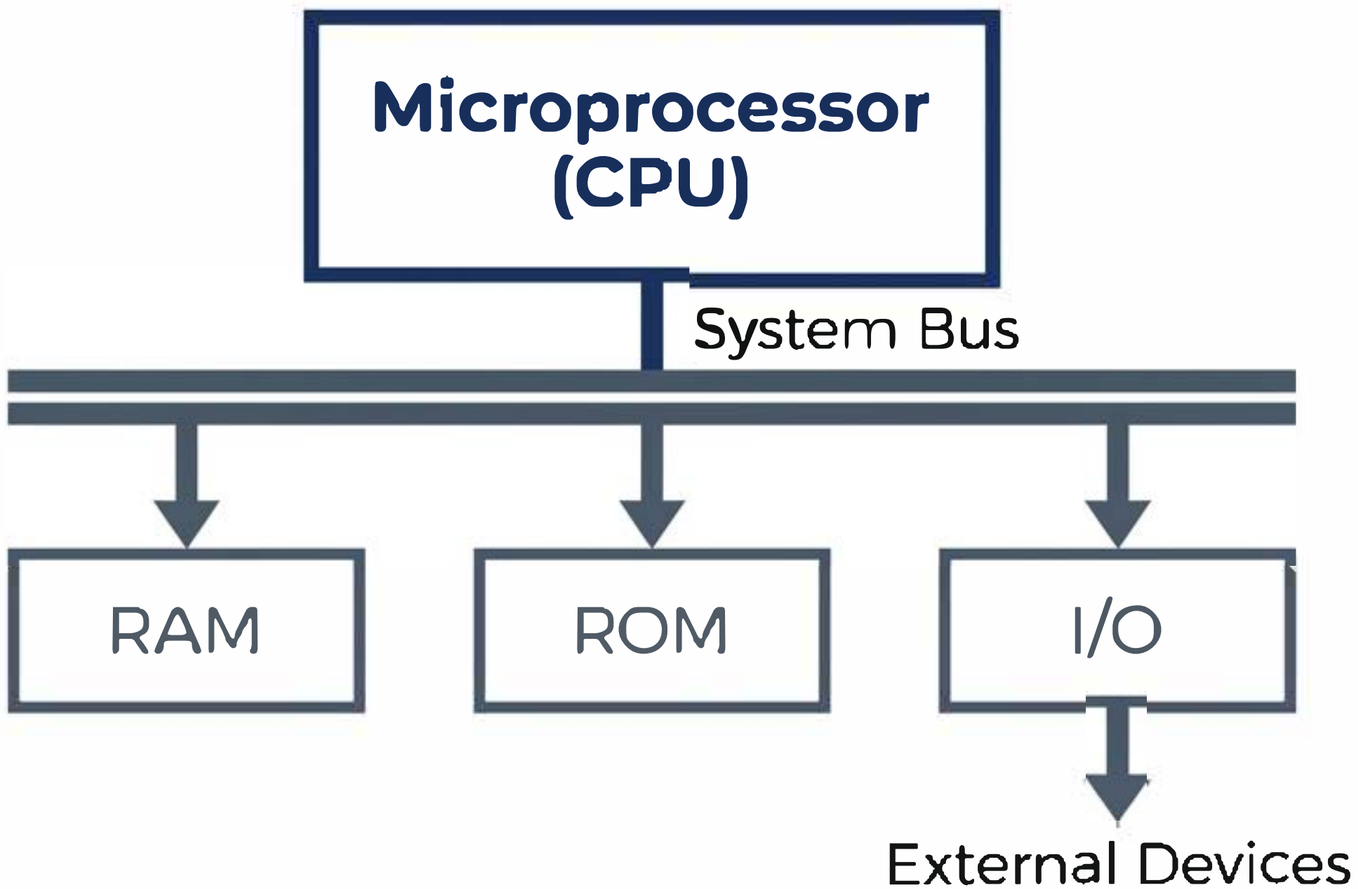
Inside the Microprocessor

Internal components are connected to the outside world via Address, Data, and Control Buses.



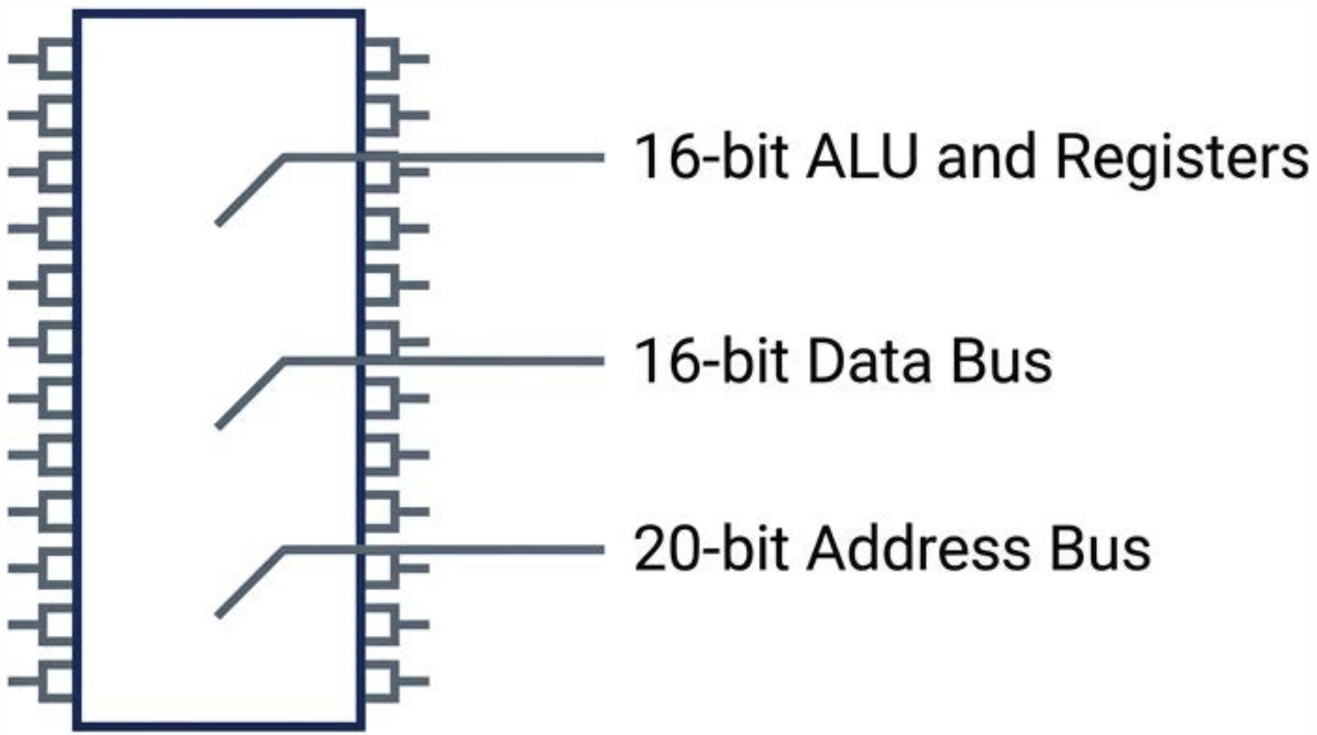
Microprocessor System Architecture

Components needed to build a complete computer reside outside the main processor chip.



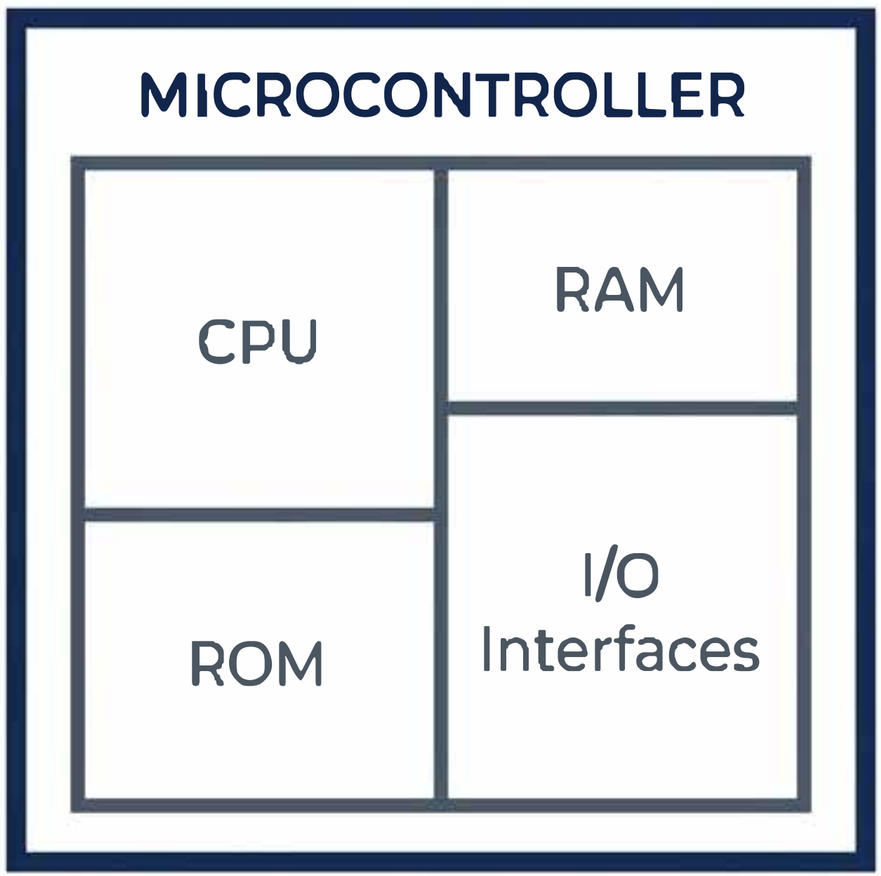
Classic Example: Intel 8086

A foundational microprocessor for learning CPU architecture. It requires external RAM, ROM, and I/O to form a complete computer.



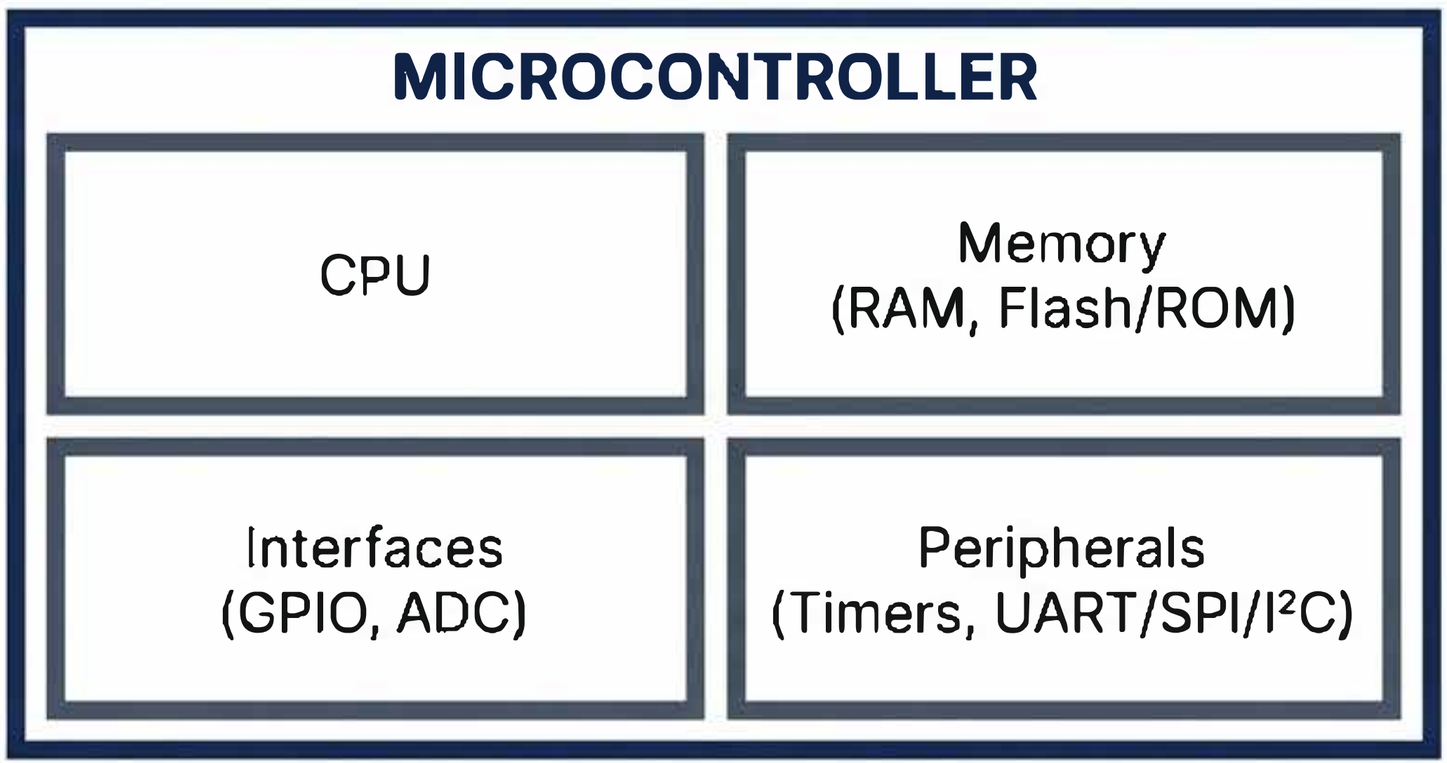
Defining the Microcontroller

A complete **small computer** integrated **into a single chip.** It is designed primarily to control specific electronic systems.



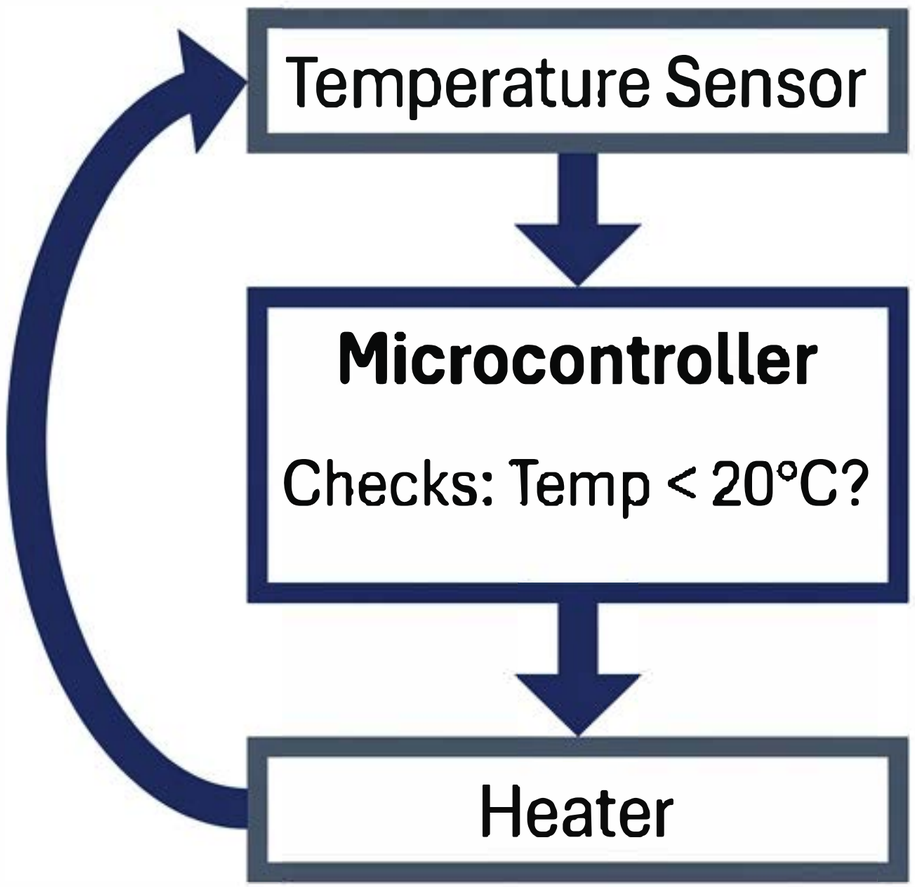
Microcontroller System Architecture

Instead of connecting many external chips, a single integrated microcontroller contains all necessary components.



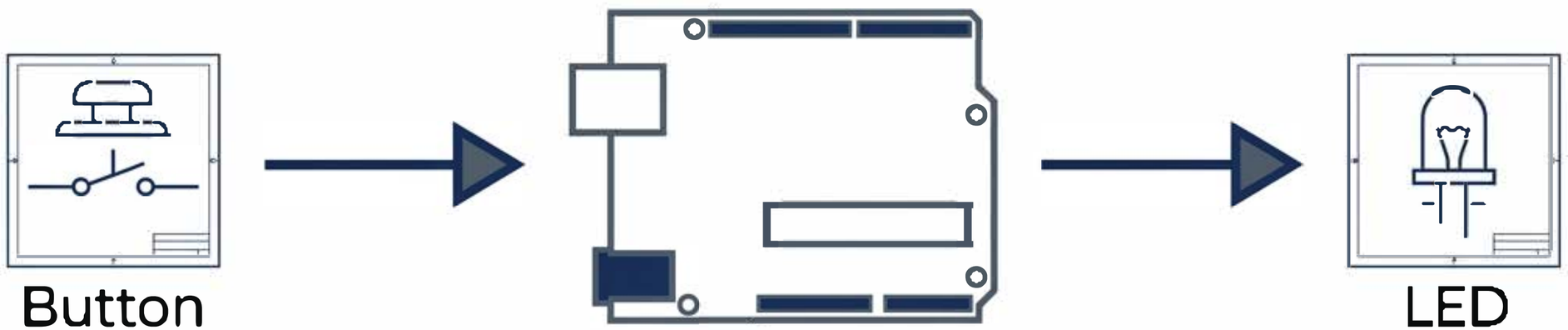
Why is it called a Controller?

A microcontroller **continuously monitors and manages** a specific, dedicated physical system.



Classroom Example: Arduino

The Arduino Uno utilizes the ATmega328P microcontroller to easily read inputs and produce outputs.



Button (Input)

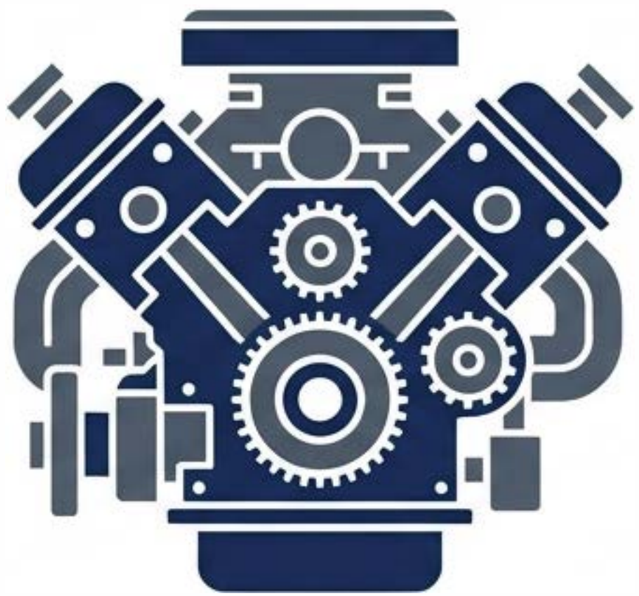
Arduino (ATmega328P)

LED (Output)

```
if (button == pressed)
  LED = ON;
else
  LED = OFF;
```

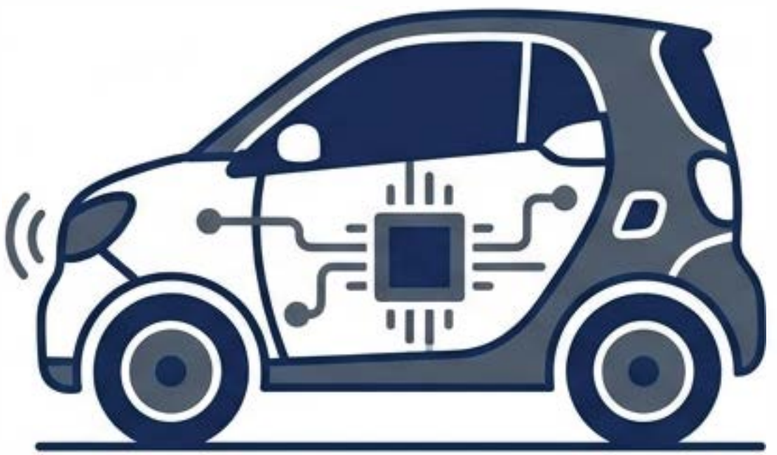
The Core Difference

Microprocessor



A powerful engine. Requires you to add a fuel system, transmission, and controls.

Microcontroller



A complete small vehicle. All necessary parts are built-in for a specific job.

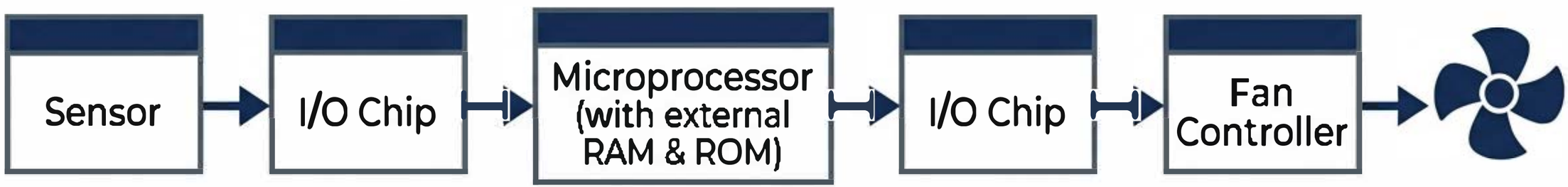
Head-to-Head Comparison

Feature	Microprocessor	Microcontroller
Main Purpose	General computing	Dedicated control
Components	Mostly external	Mostly integrated
Power & Cost	Generally higher	Generally lower
Operating System	Linux / Windows	Bare-metal / RTOS
Hardware Integration	Low	High

Practical Comparison: Building a Smart Fan

For simple control tasks, a microcontroller significantly reduces cost, size, and system complexity.

Microprocessor Approach



Microcontroller Approach



Where Do They Live?

Microprocessors

Require substantial computing capability.



Microcontrollers

Require continuous control of everyday electronic devices.



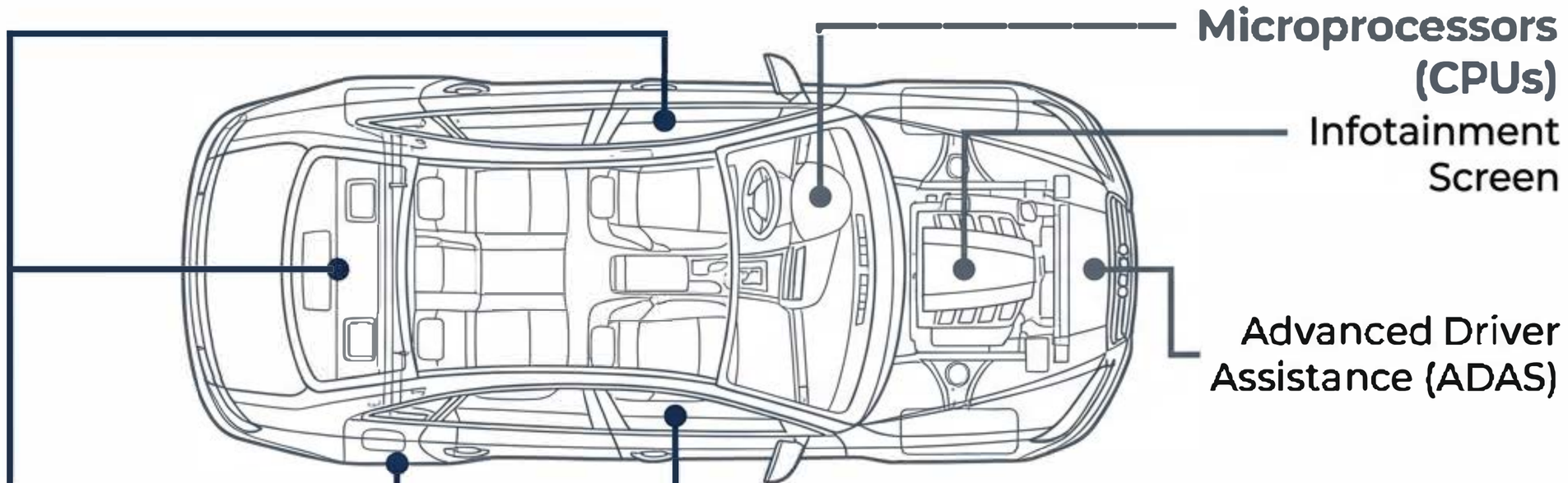
What is an Embedded System?

A computer system designed to perform a specific function as part of a larger device.

General-Purpose	Embedded
	
Many applications, powerful CPU, user installs programs.	**Specific functions, MCU, firmware** performs defined tasks, limited resources.

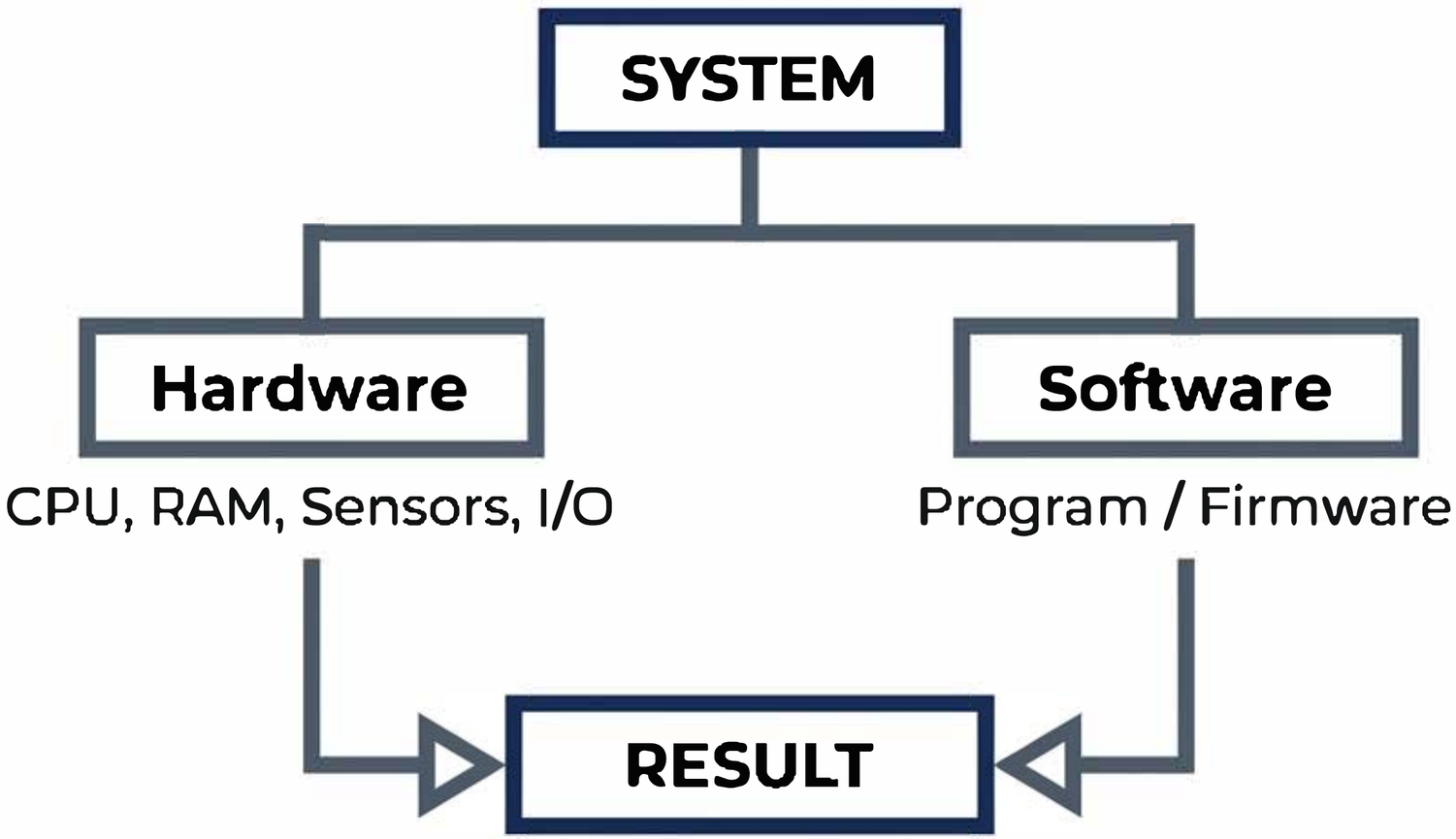
Complex Systems Use Both

Microprocessors and microcontrollers are not competitors. Modern devices utilize both for different jobs.



Hardware + Software = System

Firmware is the specific software program locked inside memory that tells an embedded device how to behave. Neither side is sufficient by itself.



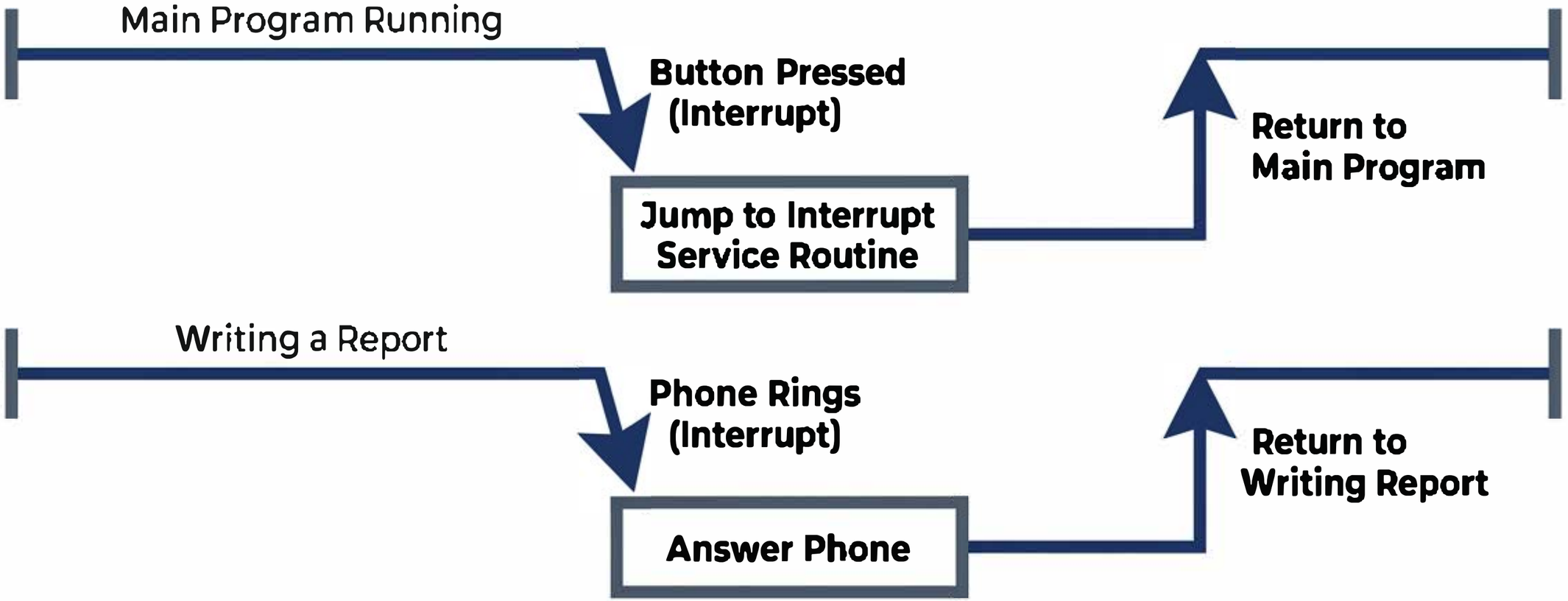
Interacting with the World

Sensors provide information to the controller. Actuators receive commands from the controller to perform physical actions.



What is an Interrupt?

A mechanism allowing a processor to temporarily stop its normal program to respond to an urgent, high-priority event.



The Golden Rule

Microprocessor = Computing. Microcontroller = Controlling.

